

Md Rayhan Ali

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Education

Rajshahi University of Engineering & Technology

Rajshahi, Bangladesh

Bachelor of Science in Computer Science and Engineering (CSE)

2019–2024

CGPA: 3.11/4.00

Relevant Coursework: Applied Statistics & Queuing Theory (Python), Neural Networks & Fuzzy Systems, Data Mining, Artificial Intelligence, Digital Image Processing, Database Systems, Parallel and Distributed Processing, Data Structures (C), Object-Oriented Programming (C++, Java), Computer Algorithms, Computer Networks, Network Security, Information System Analysis and Design.

Professional Experience

Outlier AI, Remote

Nov 2024 – Present

Independent Contractor (Contributor → Auditor)

Promoted from Contributor to Auditor for consistent task accuracy and performance. Audited and refined AI-generated responses to ensure factual correctness, clarity, and linguistic precision. Provided quality feedback to improve data annotation standards and model reliability.

Young Learners' Research Lab

Rajshahi University of Engineering & Technology (RUET), Bangladesh

May 2024 – Dec 2024

Research Assistant

Contributed to research activities involving data analysis, experiment setup, and literature review. Collaborated with faculty and student researchers to ensure methodological consistency and data integrity.

Projects

TaskLens – Full-Stack Task Management & Image Annotation Platform

Next.js, React, TypeScript,

Django, Django REST Framework, PostgreSQL, Tailwind CSS, Zustand

Developed and deployed a full-stack web application featuring JWT authentication, RESTful APIs, Kanban-style task management with drag-and-drop, image upload and polygon annotation, reusable React components, centralized state management, and responsive UI using Next.js and Django.

Snipz – Chrome Screenshot & Annotation Extension

JavaScript, Chrome Extension MV3, Canvas API

Built a lightweight Chrome extension (Manifest V3) for capturing, editing, and annotating screenshots directly in the browser. Implemented a Canvas-based editor supporting freehand drawing, lines, arrows, cropping, and resizable/rotatable annotation objects with full undo/redo history, and integrated Chrome APIs (scripting, downloads, clipboardWrite) for in-page capture and one-click download/copy.

Foundations of Neural Network Algorithms

Python, LaTeX

Implemented core neural network algorithms from scratch (KNN, Single & Multi-layer Perceptron, Kohonen SOMs, Hopfield Networks) and applied them to the Breast Cancer dataset for classification tasks.

Backpackers Home – Travel Booking Platform

HTML, CSS, JavaScript, Node.js, Express, MongoDB,

Firebase

Built a full-stack travel booking platform with a responsive frontend and a RESTful Node.js/Express backend integrated with MongoDB, enabling users to browse packages, book trips, and manage bookings through a personal dashboard. Implemented Firebase authentication, a light/dark theme toggle, and animated UI components

(AOS, Swiper.js).

Meal Management System

Java, Java Swing, MySQL, NetBeans

Developed a Java Swing desktop application to automate meal tracking, expense calculation, and billing for a student hostel mess, backed by a MySQL database. Implemented secure admin authentication and a monthly closing module that automatically computes per-member totals, reducing manual calculation errors.

Technical Skills and Interests

Programming: Python, C, C++, Java, Matlab, Assembly
Research Areas: Deep Learning, Computer Vision, Cybersecurity
Data Science: DL Architectures (Transformer, CNN, U-Net, GAN), Retrieval-Augmented Generation (RAG)
Frameworks: PyTorch, HuggingFace, TensorFlow, Keras, Android Studio, LaTeX
Web: Django, JavaScript, HTML, CSS, MySQL, PostgreSQL, Firebase

Publications

ICCIT 2024 (Published): A Hybrid 3D-2D Convolutional Neural Network with Spatial Attention for Enhanced Spectral-Spatial Classification of Hyperspectral Images (Undergrad Thesis)
Designed and implemented a hybrid 3D-2D Convolutional Neural Network (CNN) architecture with a spatial attention mechanism for hyperspectral image (HSI) classification. The model utilizes Z-score normalization and Incremental Principal Component Analysis (IPCA) for efficient dimensionality reduction, optimizing computational resources, and enhancing spectral-spatial feature extraction.
DOI: 10.1109/ICCIT64611.2024.11022565

Standardized Test Scores

GRE: Overall: 313
IELTS: Overall: 7.5

Verbal: 149, Quantitative: 164, Analytical Writing: 3.0 (Oct 2024)
Listening: 9.0, Reading: 7.5, Writing: 6.5, Speaking: 6.0 (Nov 2024)

References

Sadia Zaman Mishu: Assistant Professor, Dept. of CSE, RUET
Md. Azmain Yakin Srizon: Assistant Professor, Dept. of CSE, RUET

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